DISTINGUISHED TRIANGLE ASSOCIATED TO AN OPEN COMPLEMENT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Short exact sequence associated to an open (Etale Cohomology, Lemma etale-cohomology-lemma-ses-associated-to-open, canonical tag 095L). cited at proof lines 2657-2658. The sheaf-level triangle the derived statement follows from.

Source reference. Original lemma 095L: sources/stacks-chow/tags.txt, source lines 11589–11589. Cited at proof lines 2657-2658: the sheaf-level triangle whose derived version is the statement.

Lemma 0.2. *Let X be a scheme. Let $Z \subset X$ be a closed subscheme and let $U \subset X$ be the complement. Denote $i : Z \to X$ and $j : U \to X$ the inclusion morphisms. Let Λ be a ring. Let $K \in D(X_{étale}, \Lambda)$. There is a distinguished triangle*

$$j_!j^{-1}K \to K \to i_*i^{-1}K \to j_!j^{-1}K[1]$$

in $D(X_{étale}, \Lambda)$.

Proof. Immediate consequence of Étale Cohomology, Lemma 095L and the fact that the functors $j_!$, j^{-1}, i_*, i^{-1} are exact and hence their derived versions are computed by applying these functors to any complex of sheaves representing K. □

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